C2 - Contact-Event Phase Synchronisation Detailed Flow

Reference / event preparation

Retain the complete C1 FIR-QP controller
h, S, AR, q/r/slew, G
the pitch-recovery law is unchanged

Load scheduled contact transitions from the FDDP reference
left / right touchdown and liftoff
$t_{event,ref}$

Online phase alignment + C1 correction

Toe/heel contact observer
sensor values → site active → leg active
observed events: {leg, kind, t_{obs}}

Match observed and expected events inside the time window
$\varepsilon_t = t_{obs} - t_{expected}$
matched / skipped / unmatched

Compute the bounded phase-offset update
$\Delta t_{target} = clip(t_{expected} - t_{obs}, \pm\Delta t_{max})$
$\Delta t_\varphi{}^{+} = clip(\Delta t_\varphi + clip(K_\varphi(\Delta t_{target} - \Delta t_\varphi), \pm\delta t_{step}), \pm\Delta t_{max})$

Query the phase-aligned reference time
$t_\varphi = clip(t + \Delta t_\varphi, 0, T)$

Interpolate the nominal reference at phase time
$q_{nom} = q_{ref}(t_\varphi)$
$\theta_{ref} = \theta_{ref}(t_\varphi)$

Run the unchanged C1 FIR-QP correction
$e_k \rightarrow E_{free} + SA \rightarrow A^* \rightarrow \alpha_k$

Form the runtime position command
$q_{cmd} = q_{ref}(t_\varphi) + G\alpha_k$

Log phase / contact / pitch / QP evidence
matched and skipped events / timing error / phase offset
contacts / pitch metrics / correction / solve time

Current evidence boundary: the double-support CoM-shift reference contains no scheduled contact transition. Contact monitoring, event-capable implementation, and C1 compatibility are supported; walking-gait phase synchronisation is not validated.